



## Generating a sequence using a term-to-term rule

1 Which of these best describes what a term in a sequence is? Tick **1** correct answer

- ☐ The number of patterns in a sequence
- ☐ The first number in a sequence
- ☐ The multiplier to get from one value to the next
- ☐ The difference between successive values
- ☒ An individual diagram or value in a sequence

2 What would be the next value in this sequence if it has a constant additive rule? -1, 6, 13, ... Fill in the blank

3 What would be the next value in this sequence if it has a constant multiplicative rule? -1, 6, -36, ... Fill in the blank

4 Match the next 2 numbers in the sequence which starts 16, 4, ... with the rule the sequence is following. Write the correct letter in each box

|   |                   |
|---|-------------------|
| a | ..., 1, 0.25, ... |
| b | ..., -8, -20, ... |
| c | ..., -2, -5, ...  |
| d | ..., -6, -14, ... |
| e | ..., 2, 10, ...   |

|   |                                                    |
|---|----------------------------------------------------|
| e | subtract 12, then subtract 2, then subtract -8 ... |
| d | subtract 12, then subtract 10, then subtract 8 ... |
| a | Multiply the previous term by $\frac{1}{4}$        |
| c | halve the previous term and subtract 4             |
| b | add (-12) to the previous term                     |

5 Calculate the number halfway between 16 and 34 Fill in the blank

25



6 Match these calculations to the correct answers. Write the correct letter in each box

|   |                  |
|---|------------------|
| a | $4.1 \times 0.1$ |
| b | $4.1 \div 0.1$   |
| c | $41 \times 10$   |
| d | $41 \times 0.1$  |
| e | $410 \div 0.1$   |

|          |      |
|----------|------|
| <i>b</i> | 41   |
| <i>c</i> | 410  |
| <i>a</i> | 0.41 |
| <i>e</i> | 4100 |
| <i>d</i> | 4.1  |

